

0902.0431: residuals

equations measured 2026-09-11T09:58:26Z against the model of 2026-09-10

Corrected (10)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E00425 (was)	70	■ 0.082	$\begin{aligned} & A \times A^* \times Y A^* = A(X \times Y) A^* \\ & \left(P M A^* \right)^* \left(P N A^* \right) = A M^* P^* P N A = A(M^* N) A, \\ & \left(P M A^* \right) (A Y A^*) = P(M Y) A^*, \\ & \overline{P M A^* \times P N A^*} = \overline{{}^t \tilde{P}(M \times N) {}^t \tilde{A}^*} = \overline{P M \times N} A^*, \text{ etc. } \end{aligned}$	$\begin{aligned} & A X A^* \times A Y A^* = A(X \times Y) A^* \\ & (P M A^*)^* (P N A^*) = A M^* P^* P N A = A(M^* N) A, \\ & (P M A^*) (A Y A^*) = P(M Y) A^*, \\ & \overline{P M A^* \times P N A^*} = \overline{{}^t \tilde{P}(M \times N) {}^t \tilde{A}^*} = \overline{P M \times N} A^*, \text{ etc. } \end{aligned}$	$\begin{aligned} & A X A^* \times A Y A^* = A(X \times Y) A^*, \\ & (P M A^*)^* (P N A^*) = A M^* P^* P N A = A(M^* N) A, \\ & (P M A^*) (A Y A^*) = P(M Y) A^*, \\ & \overline{P M A^* \times P N A^*} = \overline{{}^t \tilde{P}(M \times N) {}^t \tilde{A}^*} = \overline{P M \times N} A^*, \text{ etc. } \end{aligned}$
0902.0431_E00425 (now)	70	—	$\begin{aligned} & A \times A^* \times Y A^* = A(X \times Y) A^* \\ & \left(P M A^* \right)^* \left(P N A^* \right) = A M^* P^* P N A = A(M^* N) A, \\ & \left(P M A^* \right) (A Y A^*) = P(M Y) A^*, \\ & \overline{P M A^* \times P N A^*} = \overline{{}^t \tilde{P}(M \times N) {}^t \tilde{A}^*} = \overline{P M \times N} A^*, \text{ etc. } \end{aligned}$	$\begin{aligned} & A X A^* \times A Y A^* = A(X \times Y) A^* \\ & (P M A^*)^* (P N A^*) = A M^* P^* P N A = A(M^* N) A, \\ & (P M A^*) (A Y A^*) = P(M Y) A^*, \\ & \overline{P M A^* \times P N A^*} = \overline{{}^t \tilde{P}(M \times N) {}^t \tilde{A}^*} = \overline{P M \times N} A^*, \text{ etc. } \end{aligned}$	$\begin{aligned} & A X A^* \times A Y A^* = A(X \times Y) A^*, \\ & (P M A^*)^* (P N A^*) = A M^* P^* P N A = A(M^* N) A, \\ & (P M A^*) (A Y A^*) = P(M Y) A^*, \\ & \overline{P M A^* \times P N A^*} = \overline{{}^t \tilde{P}(M \times N) {}^t \tilde{A}^*} = \overline{P M \times N} A^*, \text{ etc. } \end{aligned}$
basis: inferred / ink					
0902.0431_E00425 (was)	70	■ 0.082	$\begin{aligned} & A \times A^* \times Y A^* = A(X \times Y) A^* \\ & \left(P M A^* \right)^* \left(P N A^* \right) = A M^* P^* P N A = A(M^* N) A, \\ & \left(P M A^* \right) (A Y A^*) = P(M Y) A^*, \\ & \overline{P M A^* \times P N A^*} = \overline{{}^t \tilde{P}(M \times N) {}^t \tilde{A}^*} = \overline{P M \times N} A^*, \text{ etc. } \end{aligned}$	$\begin{aligned} & A X A^* \times A Y A^* = A(X \times Y) A^* \\ & (P M A^*)^* (P N A^*) = A M^* P^* P N A = A(M^* N) A, \\ & (P M A^*) (A Y A^*) = P(M Y) A^*, \\ & \overline{P M A^* \times P N A^*} = \overline{{}^t \tilde{P}(M \times N) {}^t \tilde{A}^*} = \overline{P M \times N} A^*, \text{ etc. } \end{aligned}$	$\begin{aligned} & A X A^* \times A Y A^* = A(X \times Y) A^*, \\ & (P M A^*)^* (P N A^*) = A M^* P^* P N A = A(M^* N) A, \\ & (P M A^*) (A Y A^*) = P(M Y) A^*, \\ & \overline{P M A^* \times P N A^*} = \overline{{}^t \tilde{P}(M \times N) {}^t \tilde{A}^*} = \overline{P M \times N} A^*, \text{ etc. } \end{aligned}$
0902.0431_E00425 (now)	70	—	$\begin{aligned} & \nu \sum_{k=0}^3 \lambda_k H_k \\ & \frac{1}{2} \left(\lambda_0 (-H_0 + H_1 - H_2 + H_3) + \lambda_1 (-H_0 + H_1 + H_2 - \right. \\ & \quad \left. + \lambda_2 (H_0 + H_1 + H_2 + H_3) + \lambda_3 (-H_0 - H_1 + H_2 + H_3) \right) \end{aligned}$	$\begin{aligned} & = \nu \left(\sum_{k=0}^3 \lambda_k H_k \right) \\ & = \frac{1}{2} \left(\lambda_0 (-H_0 + H_1 - H_2 + H_3) + \lambda_1 (-H_0 + H_1 + H_2 - \right. \\ & \quad \left. + \lambda_2 (H_0 + H_1 + H_2 + H_3) + \lambda_3 (-H_0 - H_1 + H_2 + H_3) \right) \end{aligned}$	$\begin{aligned} & A X A^* \times A Y A^* = A(X \times Y) A^*, \\ & (P M A^*)^* (P N A^*) = A M^* P^* P N A = A(M^* N) A, \\ & (P M A^*) (A Y A^*) = P(M Y) A^*, \\ & \overline{P M A^* \times P N A^*} = \overline{{}^t \tilde{P}(M \times N) {}^t \tilde{A}^*} = \overline{P M \times N} A^*, \text{ etc. } \end{aligned}$
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value					
LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E00515 (was)	85	■ 0.002	$\begin{aligned} & \lambda_0 - \lambda_1 = 1 \quad \lambda_0 - \lambda_2 = 1 \quad \lambda_0 - \lambda_3 = 1 \\ & \lambda_1 - \lambda_2 = 0 \quad \lambda_1 - \lambda_3 = 0 \quad \lambda_2 - \lambda_3 = 0 \end{aligned}$	$\begin{aligned} \lambda_0 - \lambda_1 &= 1 & 0 & 0 & 0 & 0 & 0 \\ \lambda_0 - \lambda_2 &= 1 & 1 & 0 & 0 & 0 & 0 \\ \lambda_0 - \lambda_3 &= 1 & 1 & 1 & 0 & 0 & 2 \end{aligned}$	$\begin{aligned} \lambda_0 + \lambda_1 &= 1 & 2 & 2 & 1 & 1 & 1 \\ \lambda_0 + \lambda_2 &= 1 & 1 & 2 & 1 & 1 & 1 \\ \lambda_0 + \lambda_3 &= 1 & 1 & 1 & 1 & 1 & 1 \\ \lambda_1 + \lambda_2 &= 0 & 1 & 2 & 1 & 1 & 1 \\ \lambda_1 + \lambda_3 &= 0 & 1 & 1 & 1 & 1 & 1 \\ \lambda_2 + \lambda_3 &= 0 & 0 & 1 & 1 & 1 & 1 \end{aligned}$
0902.0431_E00515 (now)	85	—	$\begin{aligned} & \lambda_0 - \lambda_1 = 1 \quad \lambda_0 - \lambda_2 = 1 \quad \lambda_0 - \lambda_3 = 1 \\ & \lambda_1 - \lambda_2 = 0 \quad \lambda_1 - \lambda_3 = 0 \quad \lambda_2 - \lambda_3 = 0 \end{aligned}$	$\begin{aligned} \lambda_0 - \lambda_1 &= 1 & 0 & 0 & 0 & 0 & 0 \\ \lambda_0 - \lambda_2 &= 1 & 1 & 0 & 0 & 0 & 0 \\ \lambda_0 - \lambda_3 &= 1 & 1 & 1 & 0 & 0 & 0 \\ \lambda_1 - \lambda_2 &= 0 & 1 & 0 & 0 & 0 & 0 \\ \lambda_1 - \lambda_3 &= 0 & 1 & 1 & 0 & 0 & 0 \\ \lambda_2 - \lambda_3 &= 0 & 0 & 1 & 0 & 0 & 0 \end{aligned}$	$\begin{aligned} \lambda_0 + \lambda_1 &= 1 & 2 & 2 & 1 & 1 & 1 \\ \lambda_0 + \lambda_2 &= 1 & 1 & 2 & 1 & 1 & 1 \\ \lambda_0 + \lambda_3 &= 1 & 1 & 1 & 1 & 1 & 1 \\ \lambda_1 + \lambda_2 &= 0 & 1 & 2 & 1 & 1 & 1 \\ \lambda_1 + \lambda_3 &= 0 & 1 & 1 & 1 & 1 & 1 \\ \lambda_2 + \lambda_3 &= 0 & 0 & 1 & 1 & 1 & 1 \end{aligned}$
basis: inferred / ink					
0902.0431_E00515 (was)	85	■ 0.002	$\begin{aligned} & \lambda_0 - \lambda_1 = 1 \quad \lambda_0 - \lambda_2 = 1 \quad \lambda_0 - \lambda_3 = 1 \\ & \lambda_1 - \lambda_2 = 0 \quad \lambda_1 - \lambda_3 = 0 \quad \lambda_2 - \lambda_3 = 0 \end{aligned}$	$\begin{aligned} \lambda_0 - \lambda_1 &= 1 & 0 & 0 & 0 & 0 & 0 \\ \lambda_0 - \lambda_2 &= 1 & 1 & 0 & 0 & 0 & 0 \\ \lambda_0 - \lambda_3 &= 1 & 1 & 1 & 0 & 0 & 2 \end{aligned}$	$\begin{aligned} \lambda_0 + \lambda_1 &= 1 & 2 & 2 & 1 & 1 & 1 \\ \lambda_0 + \lambda_2 &= 1 & 1 & 2 & 1 & 1 & 1 \\ \lambda_0 + \lambda_3 &= 1 & 1 & 1 & 1 & 1 & 1 \\ \lambda_1 + \lambda_2 &= 0 & 1 & 2 & 1 & 1 & 1 \\ \lambda_1 + \lambda_3 &= 0 & 1 & 1 & 1 & 1 & 1 \\ \lambda_2 + \lambda_3 &= 0 & 0 & 1 & 1 & 1 & 1 \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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0902.0431_E00515 (now)	85	—	$\begin{aligned} \nu H &= \nu \bigg(\sum_{k=0}^3 \lambda_k H_k \bigg) \\ &= \frac{1}{2} (\lambda_0 (-H_0 + H_1 - H_2 + H_3) + \lambda_1 (-H_0 + H_1 + H_2 - H_3) \\ &\quad + \lambda_2 (H_0 + H_1 + H_2 + H_3) + \lambda_3 (-H_0 - H_1 + H_2 + H_3)) \\ &= \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) H_0 + \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) H_1 \end{aligned}$	$\lambda_0 - \lambda_1 = 1$ $\lambda_0 - \lambda_2 = 1$ $\lambda_0 - \lambda_3 = 1$ $\lambda_1 - \lambda_2 = 0$ $\lambda_1 - \lambda_3 = 0$ $\lambda_2 - \lambda_3 = 0$ $\lambda_0 + \lambda_1 = 1$ $\lambda_0 + \lambda_2 = 1$ $\lambda_0 + \lambda_3 = 1$ $\lambda_1 + \lambda_2 = 0$ $\lambda_1 + \lambda_3 = 0$ $\lambda_2 + \lambda_3 = 0$	
basis: inferred / ink					
0902.0431_E00756 (was)	125	■ 0.119	$\begin{array}{l} \lambda_{\{0\}} - \lambda_{\{1\}} = 1 \\ \& \lambda_{\{0\}} \& \lambda_{\{1\}} = 1 \\ 2 \& 2 \& 2 \& 2 \& 1 \\ \lambda_{\{0\}} - \lambda_{\{2\}} = 1 \\ \& \lambda_{\{0\}} \& \lambda_{\{2\}} = 1 \\ 2 \& 2 \& 1 \\ \lambda_{\{0\}} - \lambda_{\{3\}} = 1 \\ \& \lambda_{\{0\}} \& \lambda_{\{3\}} = 1 \\ 1 \& 2 \& 2 \& 2 \& 1 \\ \lambda_{\{1\}} - \lambda_{\{2\}} = 0 \\ \& \lambda_{\{1\}} \& \lambda_{\{2\}} = 0 \\ 1 \& 2 \& 2 \& 2 \& 1 \\ \lambda_{\{1\}} - \lambda_{\{3\}} = 0 \\ \& \lambda_{\{1\}} \& \lambda_{\{3\}} = 0 \\ 1 \& 2 \& 2 \& 1 \\ \lambda_{\{2\}} - \lambda_{\{3\}} = 0 \\ \& \lambda_{\{2\}} \& \lambda_{\{3\}} = 0 \\ 2 \& 2 \& 1 \\ \lambda_{\{0\}} + \frac{1}{2} (\mu_2 - \mu_3) = 1 \\ \& \lambda_{\{0\}} + \frac{1}{2} (\mu_2 - \mu_3) = 0 \\ \lambda_2 + \frac{1}{2} (\mu_2 - \mu_3) = 0 \\ + \frac{1}{2} (\mu_2 - \mu_3) = 0 \end{array}$	$\lambda_0 - \lambda_1 = 1$ $\lambda_0 - \lambda_2 = 1$ $\lambda_0 - \lambda_3 = 1$ $\lambda_1 - \lambda_2 = 0$ $\lambda_1 - \lambda_3 = 0$ $\lambda_2 - \lambda_3 = 0$ $\lambda_0 + \lambda_1 = 1$ $\lambda_0 + \lambda_2 = 1$ $\lambda_0 + \lambda_3 = 1$ $\lambda_1 + \lambda_2 = 0$ $\lambda_1 + \lambda_3 = 0$ $\lambda_2 + \lambda_3 = 0$	$\lambda_0 - \lambda_1 = 1$ $\lambda_0 - \lambda_2 = 1$ $\lambda_0 - \lambda_3 = 1$ $\lambda_1 - \lambda_2 = 0$ $\lambda_1 - \lambda_3 = 0$ $\lambda_2 - \lambda_3 = 0$ $\lambda_0 + \lambda_1 = 1$ $\lambda_0 + \lambda_2 = 1$ $\lambda_0 + \lambda_3 = 1$ $\lambda_1 + \lambda_2 = 0$ $\lambda_1 + \lambda_3 = 0$ $\lambda_2 + \lambda_3 = 0$ $\lambda_0 + \frac{1}{2} (\mu_2 - \mu_3) = 1$ $\lambda_1 + \frac{1}{2} (\mu_2 - \mu_3) = 0$ $\lambda_2 + \frac{1}{2} (\mu_2 - \mu_3) = 0$ $\lambda_3 + \frac{1}{2} (\mu_2 - \mu_3) = 0$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

[illegible]

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
9902.0431_EQ0922 (was)	153	■ 0.102	$f^{-1}\left(\sum_{i<j<k}x_{ijk}\mathbf{e}_i\wedge\mathbf{e}_j\wedge\mathbf{e}_k\right)=\left(\begin{array}{ccc}h(x_{156}) & h(x_{164},\overline{x}_{256}) & h(x_{145},\overline{x}_{356}) \\ h(x_{256},\overline{x}_{164}) & h(x_{264}) & h(x_{245},\overline{x}_{364}) \\ h(x_{356},\overline{x}_{145}) & h(x_{364},\overline{x}_{245}) & h(x_{345}) \\ h(x_{423}) & h(x_{431},\overline{x}_{523}) & h(x_{412},\overline{x}_{623}) \\ h(x_{523},\overline{x}_{431}) & h(x_{531}) & h(x_{512},\overline{x}_{631}) \\ h(x_{623},\overline{x}_{412}) & h(x_{631},\overline{x}_{512}) & h(x_{612})\end{array}\right),$	$f^{-1}\left(\sum_{i<j<k}x_{ijk}\mathbf{e}_i\wedge\mathbf{e}_j\wedge\mathbf{e}_k\right)=\left(\begin{array}{ccc}h(x_{156}) & h(x_{164},\overline{x}_{256}) & h(x_{145},\overline{x}_{356}) \\ h(x_{256},\overline{x}_{164}) & h(x_{264}) & h(x_{245},\overline{x}_{364}) \\ h(x_{356},\overline{x}_{145}) & h(x_{364},\overline{x}_{245}) & h(x_{345}) \\ h(x_{423}) & h(x_{431},\overline{x}_{523}) & h(x_{412},\overline{x}_{623}) \\ h(x_{523},\overline{x}_{431}) & h(x_{531}) & h(x_{512},\overline{x}_{631}) \\ h(x_{623},\overline{x}_{412}) & h(x_{631},\overline{x}_{512}) & h(x_{612})\end{array}\right),$	$f^{-1}\left(\sum_{i<j<k}x_{ijk}\mathbf{e}_i\wedge\mathbf{e}_j\wedge\mathbf{e}_k\right)=\left(\begin{array}{ccc}h(x_{156}) & h(x_{164},\overline{x}_{256}) & h(x_{145},\overline{x}_{356}) \\ h(x_{256},\overline{x}_{164}) & h(x_{264}) & h(x_{245},\overline{x}_{364}) \\ h(x_{356},\overline{x}_{145}) & h(x_{364},\overline{x}_{245}) & h(x_{345}) \\ h(x_{423}) & h(x_{431},\overline{x}_{523}) & h(x_{412},\overline{x}_{623}) \\ h(x_{523},\overline{x}_{431}) & h(x_{531}) & h(x_{512},\overline{x}_{631}) \\ h(x_{623},\overline{x}_{412}) & h(x_{631},\overline{x}_{512}) & h(x_{612})\end{array}\right),$
9902.0431_EQ0922 (now)	153	—	$f^{-1}\left(\sum_{i<j<k}x_{ijk}\mathbf{e}_i\wedge\mathbf{e}_j\wedge\mathbf{e}_k\right)=\left(\begin{array}{ccc}h(x_{156}) & h(x_{164},\overline{x}_{256}) & h(x_{145},\overline{x}_{356}) \\ h(x_{256},\overline{x}_{164}) & h(x_{264}) & h(x_{245},\overline{x}_{364}) \\ h(x_{356},\overline{x}_{145}) & h(x_{364},\overline{x}_{245}) & h(x_{345}) \\ h(x_{423}) & h(x_{431},\overline{x}_{523}) & h(x_{412},\overline{x}_{623}) \\ h(x_{523},\overline{x}_{431}) & h(x_{531}) & h(x_{512},\overline{x}_{631}) \\ h(x_{623},\overline{x}_{412}) & h(x_{631},\overline{x}_{512}) & h(x_{612})\end{array}\right),$	$f^{-1}\left(\sum_{i<j<k}x_{ijk}\mathbf{e}_i\wedge\mathbf{e}_j\wedge\mathbf{e}_k\right)=\left(\begin{array}{ccc}h(x_{156}) & h(x_{164},\overline{x}_{256}) & h(x_{145},\overline{x}_{356}) \\ h(x_{256},\overline{x}_{164}) & h(x_{264}) & h(x_{245},\overline{x}_{364}) \\ h(x_{356},\overline{x}_{145}) & h(x_{364},\overline{x}_{245}) & h(x_{345}) \\ h(x_{423}) & h(x_{431},\overline{x}_{523}) & h(x_{412},\overline{x}_{623}) \\ h(x_{523},\overline{x}_{431}) & h(x_{531}) & h(x_{512},\overline{x}_{631}) \\ h(x_{623},\overline{x}_{412}) & h(x_{631},\overline{x}_{512}) & h(x_{612})\end{array}\right),$	$f^{-1}\left(\sum_{i<j<k}x_{ijk}\mathbf{e}_i\wedge\mathbf{e}_j\wedge\mathbf{e}_k\right)=\left(\begin{array}{ccc}h(x_{156}) & h(x_{164},\overline{x}_{256}) & h(x_{145},\overline{x}_{356}) \\ h(x_{256},\overline{x}_{164}) & h(x_{264}) & h(x_{245},\overline{x}_{364}) \\ h(x_{356},\overline{x}_{145}) & h(x_{364},\overline{x}_{245}) & h(x_{345}) \\ h(x_{423}) & h(x_{431},\overline{x}_{523}) & h(x_{412},\overline{x}_{623}) \\ h(x_{523},\overline{x}_{431}) & h(x_{531}) & h(x_{512},\overline{x}_{631}) \\ h(x_{623},\overline{x}_{412}) & h(x_{631},\overline{x}_{512}) & h(x_{612})\end{array}\right),$
basis: inferred / ink Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E01032 (was)	173	■ 0.022	$\begin{aligned} & \frac{1}{2}\left(\lambda_0+\right. \\ & \left.\lambda_1+\lambda_2-\lambda_3\right)+\frac{1}{2} \\ & \mu_2-\frac{1}{3} \nu=1 \quad 2 \quad 3 \quad 3 \quad 2 \quad 2 \quad 1 \quad 1 \\ & \frac{1}{2}\left(\lambda_0+\lambda_1-\lambda_2+\lambda_3\right)+\frac{1}{2} \mu_2-\frac{2}{3} \nu=1 \quad 2 \quad 3 \quad 3 \quad 2 \quad 1 \quad 1 \quad 1 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right)+\frac{1}{2} \mu_2-\frac{2}{3} \nu=1 \quad 2 \quad 3 \quad 3 \quad 2 \quad 1 \quad 0 \quad 1 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2-\lambda_3\right)+\frac{1}{2} \mu_2-\frac{2}{3} \nu=1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \\ & \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2-\lambda_3\right)-\frac{1}{2} \mu_2+\frac{2}{3} \nu=1 \quad 2 \quad 3 \quad 2 \quad 2 \quad 2 \quad 1 \quad 2 \\ & \frac{1}{2}\left(\lambda_0+\lambda_1-\lambda_2+\lambda_3\right)-\frac{1}{2} \mu_2+\frac{2}{3} \nu=1 \quad 2 \quad 3 \quad 2 \quad 2 \quad 1 \quad 1 \quad 2 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right)-\frac{1}{2} \mu_2-\frac{2}{3} \nu=1 \quad 2 \quad 3 \quad 2 \quad 2 \quad 1 \quad 0 \quad 2 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2-\lambda_3\right)-\frac{1}{2} \mu_2+\frac{2}{3} \nu=1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \\ & \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2+\lambda_3\right)+\frac{1}{2} \mu_3-\frac{2}{3} \nu=1 \quad 2 \quad 3 \quad 3 \quad 3 \quad 2 \quad 1 \quad 1 \\ & \frac{1}{2}\left(\lambda_0+\lambda_1-\lambda_2-\lambda_3\right)+\frac{1}{2} \mu_3-\frac{2}{3} \nu=1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2-\lambda_3\right)+\frac{1}{2} \mu_3-\frac{2}{3} \nu=1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2+\lambda_3\right)+\frac{1}{2} \mu_3-\frac{2}{3} \nu=1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \\ & \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2+\lambda_3\right)-\frac{1}{2} \mu_3+\frac{2}{3} \nu=1 \quad 3 \quad 5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 3 \\ & \frac{1}{2}\left(\lambda_0+\lambda_1-\lambda_2-\lambda_3\right)-\frac{1}{2} \mu_3+\frac{2}{3} \nu=1 \quad 2 \quad 3 \quad 2 \quad 1 \quad 1 \quad 1 \quad 2 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2-\lambda_3\right)-\frac{1}{2} \mu_3+\frac{2}{3} \nu=1 \quad 2 \quad 3 \quad 2 \quad 1 \quad 1 \quad 0 \quad 2 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2+\lambda_3\right)-\frac{1}{2} \mu_3+\frac{2}{3} \nu=1 \quad 2 \quad 3 \quad 2 \quad 1 \quad 0 \quad 0 \quad 2 \end{aligned}$	(not rendered)	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E01032 (now)	173	—	$\begin{aligned} \kappa\pi H &= \tfrac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3)H_0 + \tfrac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &\quad + \tfrac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \tfrac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3, \end{aligned}$	$\kappa\pi H = \tfrac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3)H_0 + \tfrac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 + \tfrac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \tfrac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3,$	$\begin{aligned} \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 2 & 3 & 3 & 2 & 2 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 2 & 3 & 3 & 2 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 2 & 3 & 3 & 2 & 1 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}\mu_2 + \frac{2}{3}\nu &= 1 & 2 & 3 & 2 & 2 & 2 & 1 & 2 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}\mu_2 + \frac{2}{3}\nu &= 1 & 2 & 3 & 2 & 2 & 1 & 1 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 2 & 3 & 2 & 2 & 1 & 0 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}\mu_2 + \frac{2}{3}\nu &= 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 1 & 2 & 3 & 3 & 3 & 2 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}\mu_3 + \frac{2}{3}\nu &= 1 & 3 & 5 & 4 & 3 & 2 & 1 & 3 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}\mu_3 + \frac{2}{3}\nu &= 1 & 2 & 3 & 2 & 1 & 1 & 1 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}\mu_3 + \frac{2}{3}\nu &= 1 & 2 & 3 & 2 & 1 & 1 & 0 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}\mu_3 + \frac{2}{3}\nu &= 1 & 2 & 3 & 2 & 1 & 0 & 0 & 2 \end{aligned}$
basis: inferred / ink					
0902.0431_E01059 (was)	180	■ 0.093	$\begin{aligned} &\&=\left\{\alpha\in\left(E_8\right)^{\left\{1,1-,1-\right\}}\mid\tau\tilde{\lambda}\alpha=\alpha\tau\tilde{\lambda}\right\} \\ &\&=\left\{\alpha\in E_7^C\mid\tau\tilde{\lambda}\alpha=\alpha\tau\tilde{\lambda}\right\}\text{ (Proposition 5.7.1)} \\ &\&=\left\{\alpha\in E_7^C\mid\tau\lambda\alpha=\alpha\tau\lambda\right\}\text{ (by the correspondence to Proposition 5.7.1)} \\ &\&=E_7\text{ (Lemma 4.3.3.(4)).} \end{aligned}$	$\begin{aligned} &=\left\{\alpha\in\left(E_8^C\right)_{1,1-,1-}\mid\tau\tilde{\lambda}\alpha=\alpha\tau\tilde{\lambda}\right\} \\ &=\left\{\alpha\in E_7^C\mid\tau\tilde{\lambda}\alpha=\alpha\tau\tilde{\lambda}\right\}\text{ (Proposition 5.7.1)} \\ &=\left\{\alpha\in E_7^C\mid\tau\lambda\alpha=\alpha\tau\lambda\right\}\text{ (by the correspondence to Proposition 5.7.1)} \\ &=E_7\text{ (Lemma 4.3.3.(4)).} \end{aligned}$	$\begin{aligned} &=\left\{\alpha\in\left(E_8^C\right)_{1,1-,1-}\mid\tau\tilde{\lambda}\alpha=\alpha\tau\tilde{\lambda}\right\} \\ &=\left\{\alpha\in E_7^C\mid\tau\tilde{\lambda}\alpha=\alpha\tau\tilde{\lambda}\right\}\text{ (Proposition 5.7.1)} \\ &=\left\{\alpha\in E_7^C\mid\tau\lambda\alpha=\alpha\tau\lambda\right\}\text{ (by the correspondence to Proposition 5.7.1)} \\ &=E_7\text{ (Lemma 4.3.3.(4)).} \end{aligned}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E01059 (now)	180	—	$\begin{aligned} \kappa\pi H = & \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3)H_0 + \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ & + \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3, \end{aligned}$	$\kappa\pi H = \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3)H_0 + \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 + \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3,$	$\begin{aligned} &= \{\alpha \in (E_8^C)_{1,1^-,1_-} \mid \tau\widetilde{\lambda}\alpha = \alpha\tau\widetilde{\lambda}\} \\ &= \{\alpha \in E_7^C \mid \tau\widetilde{\lambda}\alpha = \alpha\tau\widetilde{\lambda}\} \text{ (Proposition 5.7.1)} \\ &= \{\alpha \in E_7^C \mid \tau\lambda\alpha = \alpha\tau\lambda\} \text{ (by the correspondence to Proposition 5.7.1)} \\ &= E_7 \text{ (Lemma 4.3.3.(4)).} \end{aligned}$
basis: inferred / ink					
0902.0431_E01117 (was)	189	■ 0.108	$\begin{aligned} &\& (\boldsymbol{X}, \boldsymbol{Y}) = \\ &\left(X_{\{1\}}, Y_{\{1\}} \right) + \left(X_{\{2\}}, Y_{\{2\}} \right) + \\ &\left(X_{\{3\}}, Y_{\{3\}} \right) \in C, \quad \& \langle \\ &\boldsymbol{X}, \boldsymbol{Y} \rangle = \left(\langle X_{\{1\}}, Y_{\{1\}} \right) + \left(\langle X_{\{2\}}, Y_{\{2\}} \right) \\ &\left(\langle X_{\{3\}}, Y_{\{3\}} \right) \in C, \quad \& \boldsymbol{X} \times \boldsymbol{Y} = \left(\begin{array}{ll} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{array} \right) \in (\mathfrak{J}^C)^3, \\ &\boldsymbol{X} \cdot \boldsymbol{Y} = \left(\begin{array}{ll} (X_1, Y_1) & (X_1, Y_2) \\ (X_2, Y_1) & (X_2, Y_2) \\ (X_3, Y_1) & (X_2, Y_3) \\ X \vee Y & (X_3, Y_3) \end{array} \right) - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ &\boldsymbol{X}Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$	$\begin{aligned} &(\boldsymbol{X}, \boldsymbol{Y}) = (X_1, Y_1) + (X_2, Y_2) + (X_3, Y_3) \in C, \\ &\langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ &\boldsymbol{X} \times \boldsymbol{Y} = \left(\begin{array}{ll} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{array} \right) \in (\mathfrak{J}^C)^3, \\ &\boldsymbol{X} \cdot \boldsymbol{Y} = \left(\begin{array}{ll} (X_1, Y_1) & (X_1, Y_2) \\ (X_2, Y_1) & (X_2, Y_2) \\ (X_3, Y_1) & (X_2, Y_3) \\ X \vee Y & (X_3, Y_3) \end{array} \right) - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ &\boldsymbol{X}Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$	$\begin{aligned} &(\boldsymbol{X}, \boldsymbol{Y}) = (X_1, Y_1) + (X_2, Y_2) + (X_3, Y_3) \in C, \\ &\langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ &\boldsymbol{X} \times \boldsymbol{Y} = \left(\begin{array}{ll} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{array} \right) \in (\mathfrak{J}^C)^3, \\ &\boldsymbol{X} \cdot \boldsymbol{Y} = \left(\begin{array}{lll} (X_1, Y_1) & (X_1, Y_2) & (X_1, Y_3) \\ (X_2, Y_1) & (X_2, Y_2) & (X_2, Y_3) \\ (X_3, Y_1) & (X_3, Y_2) & (X_3, Y_3) \end{array} \right) - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ &\boldsymbol{X} \vee \boldsymbol{Y} = X_1 \vee Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
9982.0431_E01117 (now)	189	—	$\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y}) = (X_1, Y_1) + (X_2, Y_2) + (X_3, Y_3) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ & \boldsymbol{X} \times \boldsymbol{Y} = \begin{pmatrix} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{pmatrix} \in (\mathfrak{J}^C)^3, \\ & \boldsymbol{X} \cdot \boldsymbol{Y} = \begin{pmatrix} (X_1, Y_1) & (X_1, Y_2) & (X_1, Y_3) \\ (X_2, Y_1) & (X_2, Y_2) & (X_2, Y_3) \\ (X_3, Y_1) & (X_3, Y_2) & (X_3, Y_3) \end{pmatrix} - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ & \boldsymbol{X} \vee \boldsymbol{Y} = X_1 \vee Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$	$\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y}) = (X_1, Y_1) + (X_2, Y_2) + (X_3, Y_3) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ & \boldsymbol{X} \times \boldsymbol{Y} = \begin{pmatrix} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{pmatrix} \in (\mathfrak{J}^C)^3, \\ & \boldsymbol{X} \cdot \boldsymbol{Y} = \begin{pmatrix} (X_1, Y_1) & (X_1, Y_2) & (X_1, Y_3) \\ (X_2, Y_1) & (X_2, Y_2) & (X_2, Y_3) \\ (X_3, Y_1) & (X_3, Y_2) & (X_3, Y_3) \end{pmatrix} - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ & \boldsymbol{X} \vee \boldsymbol{Y} = X_1 \vee Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$	$\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y}) = (X_1, Y_1) + (X_2, Y_2) + (X_3, Y_3) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ & \boldsymbol{X} \times \boldsymbol{Y} = \begin{pmatrix} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{pmatrix} \in (\mathfrak{J}^C)^3, \\ & \boldsymbol{X} \cdot \boldsymbol{Y} = \begin{pmatrix} (X_1, Y_1) & (X_1, Y_2) & (X_1, Y_3) \\ (X_2, Y_1) & (X_2, Y_2) & (X_2, Y_3) \\ (X_3, Y_1) & (X_3, Y_2) & (X_3, Y_3) \end{pmatrix} - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ & \boldsymbol{X} \vee \boldsymbol{Y} = X_1 \vee Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					